

[1]: `import numpy as np`

[18]: `arr = np.arange(0,10)`

[19]: `arr`

[19]: `array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])`

[21]: `print(arr+5)`

[21]: `[ 5 6 7 8 9 10 11 12 13 14]`

[22]: `print(arr**arr)`

[22]: `[ 1 1 4 27 256 3125 46656`
 `823543 16777216 387420489]`

Instead of showing fully errors, it's just showing small error/ a warning but still gives output for other elements in array

[23]: `print(arr/arr)`

[23]: `[nan 1. 1. 1. 1. 1. 1. 1. 1. 1.]`

`/tmp/ipykernel_210441/3742253236.py:1: RuntimeWarning: invalid value encountered in divide`
`print(arr/arr)`

Sum of elements in an array

[24]: `print(arr.sum())`

[24]: `45`

To get mean of an array

[25]: `print(arr.mean())`

[25]: `4.5`

To get variance of elements in an array

[26]: `print(arr.var())`

[26]: `8.25`

To get standard deviation of elements in an array

[27]: `print(arr.std())`

[27]: `2.8722813232690143`

All we did was for 1 dimensional array, now let's do for multiple and check

[32]: `arr2d = np.arange(0,25).reshape(5,5)`
`print(arr2d)`

[32]: `[[ 0 1 2 3 4]`
 `[ 5 6 7 8 9]`
`[10 11 12 13 14]`
`[15 16 17 18 19]`
`[20 21 22 23 24]]`

Let's check sum

[37]: `print(arr2d.sum())`

[37]: `300`

Variance

[34]: `print(arr2d.var())`

[34]: `52.0`

Standard deviation

[35]: `print(arr2d.std())`

[35]: `7.211102550927978`

Now if we want to get sum of only columns/ rows:

[42]: `print(arr.sum(0)) # columns = 0`

[42]: `[50 55 60 65 70]`

[43]: `print(arr.sum(1)) # rows = 1`

[43]: `[ 10 35 60 85 110]`

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